



Green Sericulture: Eco-Friendly Innovations Shaping the Future of Silk Farming



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Abstract

Sericulture, the ancient practice of silk production, has long been a cornerstone of rural economies and cultural heritage, particularly in countries like India and China. However, traditional methods often involve significant environmental challenges, including chemical pesticide use, high water consumption, waste generation and ethical concerns over silkworm treatment. This review explores the emergence of "green sericulture"-eco-friendly innovations that prioritize sustainability, biodiversity and ethical practices while maintaining or enhancing productivity. Key advancements include organic mulberry cultivation, biological pest control, automated rearing technologies, ahimsa (non-violent) silk production and circular waste management strategies. Drawing from recent literature, this article highlights how these innovations reduce ecological footprints, improve resource efficiency and support socio-economic benefits for farmers. By integrating biotechnology, IoT monitoring and circular economy principles, green sericulture offers a viable path toward a resilient and ethical silk industry. The findings underscore the potential for sericulture to align with global sustainability goals, providing luxury textiles without compromising environmental or ethical standards.

Keywords: green sericulture, sustainable silk production, eco-friendly innovations, ahimsa silk, waste management in sericulture, circular economy

Introduction

Silk, often dubbed the "queen of fabrics," has captivated humanity for over 5,000 years, originating in ancient China and spreading along the Silk Road to influence global trade and culture. Sericulture-the integrated process of cultivating mulberry trees, rearing silkworms (primarily *Bombyx mori*) and extracting silk-remains a vital agroindustry, employing millions in rural areas of India, China and other Asian countries (Altman & Farrell, 2022). It generates high-value output from low-capital input, supports women's empowerment and contributes to foreign exchange earnings. Yet, conventional sericulture faces criticism for its environmental impacts: heavy reliance on chemical fertilizers and pesticides in mulberry fields leads to soil degradation and water pollution; intensive water uses strains resources; silkworm diseases prompt antibiotic overuse; and the standard practice of boiling cocoons kills pupae, raising ethical concerns. Additionally, waste from prunings, litter and pupae often go unmanaged, contributing to pollution (Das et al., 2025).

In response, "green sericulture" has emerged as a transformative approach, emphasizing ecological responsibility alongside economic viability. Innovations span organic farming, non-violent harvesting, biotechnology and waste valorisation (Kairy, 2020). These practices not only mitigate harm but also enhance yield and quality, making sustainable silk competitive in premium markets. This article reviews key eco-friendly innovations shaping the future of silk farming, demonstrating how sericulture can evolve into a model of sustainable agriculture.

Methodology

This article is a narrative review based on a comprehensive analysis of existing peer-reviewed literature, reports and scholarly sources published between 2014 and 2025. Sources were identified through web searches using keywords such as "sustainable sericulture," "eco-friendly sericulture," "ahimsa silk," "organic mulberry cultivation," and "sericulture waste management." Priority was given to open-access journal articles, review papers and PDFs from reputable platforms (e.g., ScienceDirect, ResearchGate and specialized entomology journals). Inclusion criteria focused on studies addressing environmental impacts, innovative practices and sustainability outcomes in sericulture. Approximately 30 sources were examined, with key papers selected for their relevance to eco-innovations. Data synthesis involved thematic categorization: mulberry cultivation, silkworm rearing, ethical production, waste management and technological advancements. In-text citations follow APA style, with full references provided at the end. No primary data collection was conducted; the review aims to synthesize evidence for an accessible yet rigorous overview suitable for researchers, policymakers and industry stakeholders.

Sustainable Mulberry Cultivation: The Foundation of Green Sericulture

Mulberry (*Morus spp.*) leaves constitute the primary food for silkworms, making cultivation practices pivotal to sustainability. Traditional methods often involve synthetic fertilizers and pesticides, leading to soil erosion, biodiversity loss and chemical runoff. Organic mulberry farming addresses these issues by using natural inputs like farmyard manure (FYM), vermicompost, biofertilizers and green manures. These enhance soil health, increase leaf yield and reduce dependency on chemicals. For instance, integrated organic practices-combining composting with botanical pest repellents-have been shown to maintain productivity while promoting microbial diversity (Kairy, 2020). Water-efficient techniques, such as drip irrigation and mulching, conserve resources in water-scarce regions. Intercropping mulberry with legumes or shade-tolerant crops boosts biodiversity and provides additional income. Mulberry trees themselves contribute to carbon sequestration and soil conservation, positioning them as climate-resilient assets (Altman & Farrell, 2022). Recent studies highlight dwarf mulberry varieties and hydroponic systems for year-round leaf production, reducing land use and seasonal limitations. Overall, organic cultivation lowers the environmental footprint while yielding nutritious leaves that improve silkworm health and silk quality.

Eco-Friendly Silkworm Rearing and Disease Management

Silkworm rearing traditionally occurs in controlled but resource-intensive environments. Green innovations focus on reducing chemical inputs and optimizing conditions.

Biological control methods-using natural predators, parasitoids and pathogens-replace synthetic pesticides for managing pests like *Uzi* fly. Integrated Pest Management (IPM) combines cultural practices (e.g., hygiene), mechanical barriers and minimal interventions, preserving beneficial insects (as reviewed in advancements literature, 2024). Automated rearing systems equipped with IoT sensors monitor temperature, humidity and airflow in real-time, minimizing energy waste and disease outbreaks. AI-driven predictive models detect early signs of infection, allowing timely non-chemical interventions. Probiotics and microbial supplements enhance silkworm immunity and digestion, reducing antibiotic use. Artificial diets supplemented with nutrients offer alternatives to mulberry leaves in off-seasons, though mulberry remains preferred for superior silk. Controlled environments enable year-round rearing, stabilizing supply chains. These technologies improve survival rates (often exceeding 90%) and cocoon quality while cutting operational costs and ecological impacts (Kairy, 2020).

Ahimsa Silk: Ethical and Non-Violent Production

A major ethical critique of sericulture is the killing of silkworm pupae by boiling or stifling cocoons to preserve continuous filaments. Ahimsa (peace) silk eliminates this by allowing moths to emerge naturally, then harvesting pierced cocoons. Primarily produced from wild or semi-domesticated species like eri (*Samia ricini*), tasar (*Antheraea paphia*) and muga (*Antheraea assamensis*), ahimsa silk aligns with Jain principles of non-violence. Eri silk, spun from open-ended cocoons, is inherently non-violent as moths break the fibers upon emergence. Though yielding shorter staples and lower volumes (about one-sixth of conventional),

ahimsa silk commands premium prices and appeals to ethical consumers. Production supports rural artisans and biodiversity, as wild silk moths feed on diverse host plants like castor. Innovations in breeding aim to extend ahimsa methods to mulberry silkworms without compromising quality, blending ethics with sustainability.

Waste Management and Circular Economy in Sericulture

Sericulture generates substantial biomass waste: mulberry prunings, silkworm litter (excreta), pierced cocoons and pupae. Traditional disposal causes pollution, but green approaches transform waste into resources. Vermicomposting and microbial degradation convert litter and prunings into nutrient-rich fertilizers, closing nutrient loops. Pupae, high in protein, serve as animal feed or human food (e.g., in entomophagy). Sericin (a by-product) finds uses in cosmetics and biomedicine. Circular models recycle water in rearing units and utilize biogas from waste for energy. These practices reduce landfill burden, lower input costs and generate additional revenue (Das *et al.*, 2025). By adopting zero-waste principles, sericulture exemplifies a circular agroindustry with minimal ecological disruption (Altman & Farrell, 2022).

Technological and Biotechnological Innovations

Beyond practices, cutting-edge tools drive green sericulture:

- Genetic engineering (e.g., CRISPR) creates disease-resistant, high-yield breeds.
- Nanotechnology enhances silk properties for medical applications.
- Blockchain ensures traceability for eco-certified silk.

These amplify sustainability when paired with organic baselines.

Conclusion

Green sericulture represents a harmonious fusion of tradition and innovation, addressing environmental, ethical and economic challenges in silk farming. From organic mulberry fields to ahimsa harvesting and waste valorisation, these practices demonstrate that luxury textiles can be produced responsibly. Supported by policy incentives, farmer training and consumer demand for sustainable goods, the industry is poised for growth (Kairy, 2020; Altman & Farrell, 2022). By embracing eco-friendly innovations, sericulture not only preserves its cultural legacy but also contributes to global goals like biodiversity conservation and circular economies. Future research should focus on scaling these approaches and quantifying long-term impacts to ensure silk remains a symbol of elegance-and sustainability.

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